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Title of the Report:
Feasibility Study for Automated Fetal Health Risk Assessment System

Project Name:
Automated Fetal Health Risk Assessment System for Rural Midwifery Care

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Executive Summary

This report evaluates the feasibility of developing an **Automated Fetal Health Risk Assessment System** tailored for rural midwives in Sri Lanka. The proposed solution is a mobile-based application that utilizes **Machine Learning (Random Forest Algorithm)** to analyze Cardiotocography (CTG) data and predict fetal health risks (Normal, Suspect, Pathological) in real-time.

Key findings indicate that the project is **Technically Feasible** due to the availability of validated datasets (UCI Machine Learning Repository) and lightweight algorithms suitable for mobile deployment. **Market Feasibility** is high, addressing a critical gap in rural healthcare where VOG specialists are scarce. **Financial Feasibility** is positive as the solution relies on open-source technologies (Python, Flutter, Firebase), minimizing development costs. The study recommends proceeding with the development phase, focusing on offline capabilities and Explainable AI integration.

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The feasibility study confirms that developing an Automated Fetal Health Risk Assessment System is a viable and impactful project. By leveraging lightweight Machine Learning, the system can provide high-level diagnostic support to rural midwives using standard mobile devices. The project addresses a proven medical gap with a technically sound and cost-effective solution. 7
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Introduction

Background

In Sri Lanka's rural healthcare sector, midwives are the primary caregivers for expectant mothers. While CTG machines are available in many peripheral hospitals, the expertise required to interpret complex fetal heart rate traces is often limited to Obstetricians (VOGs), who are not always available. This delay in diagnosis can lead to preventable infant mortality and morbidity.

Objective of the Feasibility Study

To assess the technical, operational, and financial viability of developing an AI-powered mobile tool that assists midwives in interpreting CTG data instantly and accurately..

Scope of the Feasibility Study

This study covers the feasibility of implementing the Machine Learning model, mobile application integration, offline functionality, and the "Explainable AI" feature for clinical reasoning.

Problem Statement

The core problem is the lack of real-time expert interpretation of CTG traces in rural healthcare settings.

Evidence: Current practices rely on manual observation or transferring patients to major hospitals, which causes critical delays.
Significance: Misinterpretation of fetal distress signals (such as prolonged decelerations) can result in stillbirths or cerebral palsy. A tool that provides instant, expert-level triage can significantly reduce these risks.

Project Description

Proposed Solution

The proposed solution is an Intelligent Mobile Assistant for Midwives.

Key Features:

Automated Risk Prediction: Classifies fetal health into Normal, Suspect, or Pathological categories.

Explainability: Displays the reasons for the risk (e.g., "High Abnormal Short Term Variability detected").

Offline Capability: Functions without an active internet connection.

Methodology: The system uses a Random Forest Classifier trained on the UCI Fetal Health Dataset (2,126 records). The backend is built with Python (Flask) and integrated into a mobile app.

Feasibility Analysis

Technical Feasibility

- **Requirements:** Python (Scikit-learn) for AI, Flutter for Mobile App, and local storage for offline use.
- **Assessment:** The project is highly feasible. The **Random Forest algorithm** is lightweight and has already achieved a **94.6% accuracy** in initial testing. The technology stack is standard and well-supported.
- **Challenges:** Integrating the Python backend with a mobile interface for offline inference requires optimization (using tools like TensorFlow Lite or ONNX).

Market Feasibility

- **Target Audience**: Midwives and nursing staff in rural and peripheral hospitals in Sri Lanka.
- **Demand:** There is a strong demand for low-cost digital health tools that empower frontline health workers. The solution fills a gap left by expensive, hospital-based monitoring systems like "SisPorto".

Financial Feasibility

- **Cost Estimates**: Development costs are minimal as open-source tools (VS Code, Google Colab, Flutter) and free datasets are used.
- **Savings:** Early detection of fetal distress reduces the cost of treating long-term complications (e.g., NICU care) for the healthcare system.
- **ROI:** The return is measured in "social impact" and improved health outcomes rather than direct monetary profit.

Operation Feasibility

- **Implementation:** The app is designed to fit into the existing workflow of a midwife. It requires manual input of CTG parameters, which midwives are already familiar with.
- **Risks:** Resistance to technology among older staff. This can be mitigated by a simple, user-friendly UI/UX design.

Ethical and Environmental Feasibility

- **Impact:** The system acts as a "Decision Support Tool," not a replacement for doctors. The final decision remains with the medical professional.
- **Data Privacy:** Patient data will be stored locally or anonymized, complying with medical data ethics.

Risk Analysis

Risk Type	Risk Description	Likelihood	Mitigation Strategy
Technical	Model performing poorly on new/unseen data (Overfitting).	Medium	Use Cross-Validation and collect diverse test data.

Risk Type	Risk Description	Likelihood	Mitigation Strategy
Operational	Midwives entering incorrect data values.	High	Implement input validation ranges (e.g., HR must be 50-200).
Technical	Mobile app lagging due to model processing.	Low	Use optimized "Lightweight" ML models (Random Forest).

Recommendations

Based on the analysis, the project is **Feasible**.

- **Technical:** Continue with Random Forest as it offers the best balance of accuracy and performance.
- **Operational:** Focus heavily on the UI design to ensure it is intuitive for non-technical users.
- **Next Steps:** Proceed to full backend integration and develop a beta version for user testing.

Conclusion

The feasibility study confirms that developing an Automated Fetal Health Risk Assessment System is a viable and impactful project. By leveraging lightweight Machine Learning, the system can provide high-level diagnostic support to rural midwives using standard mobile devices. The project addresses a proven medical gap with a technically sound and cost-effective solution.

References

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